

Phase 9 CVRP Solver Evolution Report

Overview

- Condition count: 4.
- Best held-out condition: phase9_baseline_clarke_wright_savings.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Mean Novelty	Mean Complexity
phase9_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	42.75616	0.0	0.0
phase9_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	81.87964	0.0	0.0
phase9_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1463.2087	0.0	0.0
phase9_solver_evolution	solver_evolution	1.0	0.237296	0.237296	263.83732	0.594006	17.005

Condition Notes

phase9_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 42.75616.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
 - X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
 - X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.

- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 81.87964.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
 - X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
 - X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1463.2087.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:
 - X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
 - X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_solver_evolution

- Execution mode: solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.276613.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.237296.
- Held-out runtime ms: 263.83732.
- Robustness dispersion across held-out structure families: 0.000573.
- Worst held-out instances:
 - X-n247-k50: feasible=True, penalized gap=0.387831, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.32015, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.28469, errors=[]

- X-n190-k8: feasible=True, penalized gap=0.108422, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.085385, errors=[]

Judge Note

Summary of Phase-9 CVRP Solver Evolution Suite

Solver Condition	Feasibility Rate	Held-out Feasible Gap	Held-out Runtime (ms)	Robustness Dispersion	Complexity	Notes
Baseline Nearest Neighbor	1.0	0.2734	42.8	0.0028	0	Fastest runtime
Baseline Clarke-Wright Savings	1.0	**0.0738**	81.9	0.0078	0	Best gap (objective) among baselines
Baseline Regret Insertion LS	1.0	0.4664	1463.2	0.0869	0	Longest runtime, worst gap
Solver Evolution (evolved solver)	1.0	0.2373	263.8	**0.0006**	17.0	Moderate gap and runtime, best robustness

Conservative Evaluation

- ****Feasibility:**** All solvers maintain perfect feasibility (1.0 rate) on held-out instances.
- ****Objective Gap:**** The evolved solver's gap (0.2373) is better than nearest neighbor (0.2734) and regret insertion (0.4664) but notably worse than Clarke-Wright savings baseline (0.0738).
- ****Runtime:**** Evolved solver runs significantly slower than the fastest (nearest neighbor: 42.8 ms vs. 263.8 ms) but much faster than regret insertion (1463.2 ms).
- ****Robustness:**** Evolved solver shows the lowest robustness dispersion (0.0006), indicating more consistent performance across held-out structural families.
- ****Complexity & Novelty:**** Evolution introduced higher complexity (mean complexity = 17) and code novelty (~0.59), indicating structural changes.

Conclusion

The evolved phase-9 solver demonstrates robust and feasible performance with a reasonable trade-off in runtime and complexity. However, it ****does not clearly beat**** the best fixed baseline (Clarke-Wright savings) in terms of objective gap on held-out structure families, though it offers better robustness and intermediate runtime.